A4: Integrated Layout Design of an Integer-N Phase Locked Loop

EN4431: Analog IC Design

May 8, 2026

Recall the group-wise layout blocks that were designed for the previous assignment (A3).

- Group 01: Phase frequency detector, Charge pump, and Loop filter
- Group 02: Current-starved ring VCO
- Group 03: Programmable divider

Now you need to use the above three layout designs of each block for the next stage. In this stage, each student needs to complete the integrated layout design individually by considering them as hard IPs. You may also use the design "A Ring-Oscillator-Based 2.4 GHz Integer-N PLL Design in SkyWater 130nm Open-Source Technology - ISPACS 2025" [1] as a reference.

Choose an appropriate floorplan in the top-level design such that the design is made as small as possible so it can be incorporated into future Tiny Tapeout runs with minimal area overhead (less than 160 μm $\times$ 225 μm). Also, organize your design files using the folder structure available in Submission template for Tiny Tapeout SKY130.

Instructions

Please submit your report including the following information;

- Justifications for placement and design choices.
- Layout images and physical dimensions of the design.
- DRC and LVS results.

Due Date: 21st June 2026 by 11.59 PM
Target Tapeout Run: TTSKY26c (September 2026) [Link]

References

[1] N. F. Assaify, A. N. Irfansyah, and T. Mujiono, "A ring-oscillator-based 2.4 ghz integer-n pll design in skywater 130nm open-source technology," in *2025 International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, 2025, pp. 1–5.